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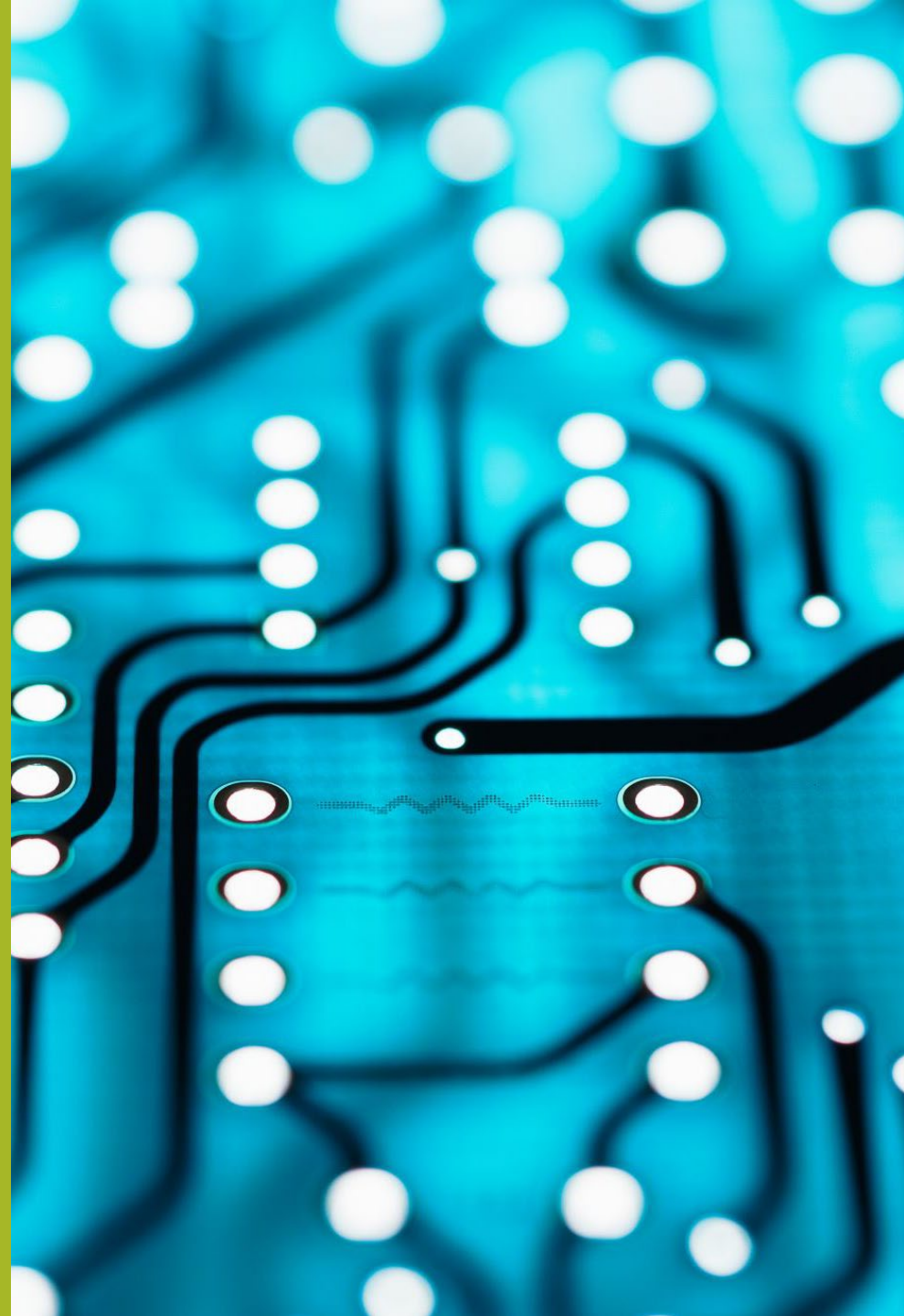
PHYC10004 Physics 2 Electromagnetism

Part 3

Electric circuits



Course Outline

Finished

Finished

1. Electrostatics
   - Stationary charged particles
   - Electrostatic forces and fields
   - Gauss' Law
2. Energy and potential
   - Electric potential energy
   - Electric potential
   - Capacitance
3. Circuit Theory
   - Electrons moving in wires
   - Resistors, capacitors, batteries
4. Magnetism
   - Moving charged particles
   - Magnetic fields, forces, materials
   - Magnetic induction

Part 3 – Outline

~ 2 lectures

- Chapters 26-27 - Textbook
- Conductors and insulators
- Microscopic view of electric current
- Power and electric circuits
- Kirchhoff's circuit rules
- Circuits with capacitors

Part 3
Electric current

Key concepts

THIS TOPIC

- Microscopic origins of electric current
- Current and current density
- Resistance
- Ohm’s Law

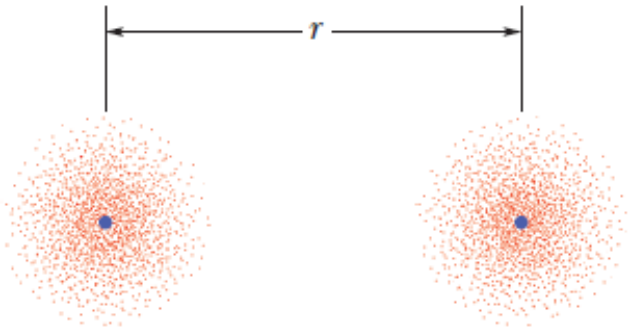
The Atom

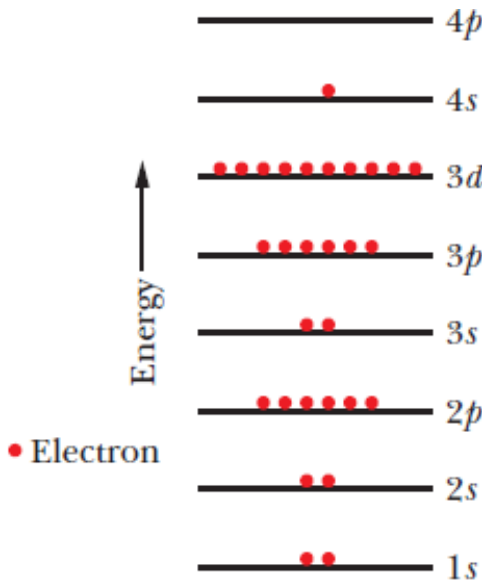
Atoms distant from each other

Electrons arranged in shells

Filled from the lowest levels

This pattern determines the structure of the Periodic Table





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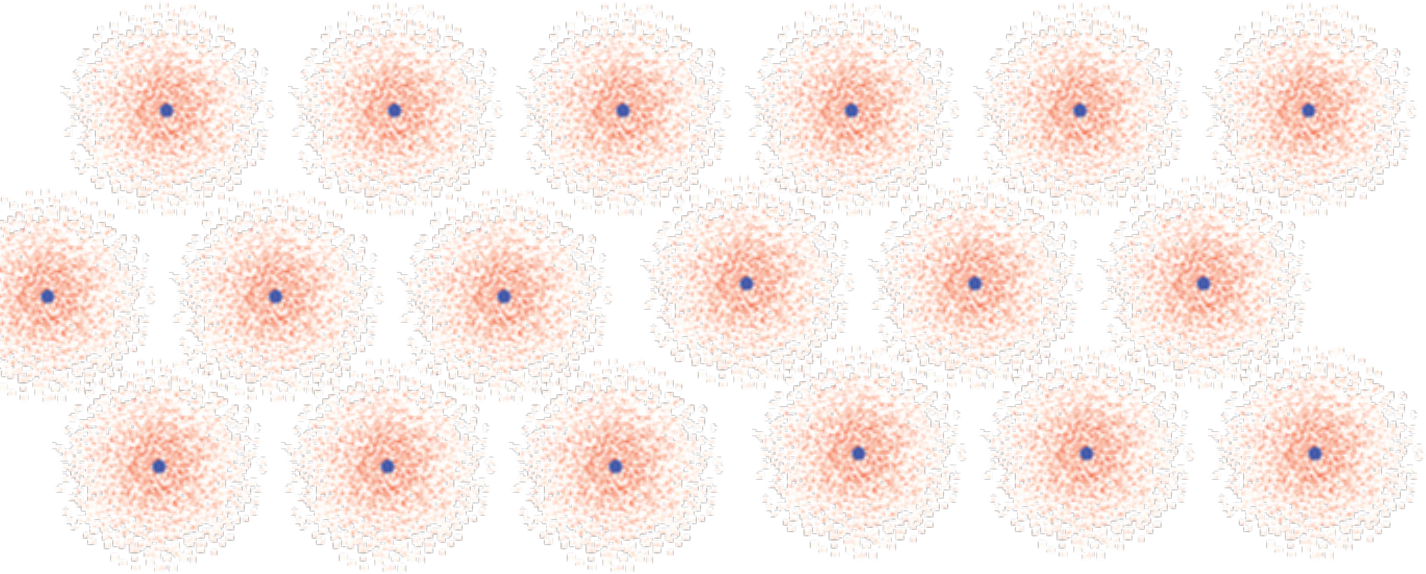
Ideal crystalline solid

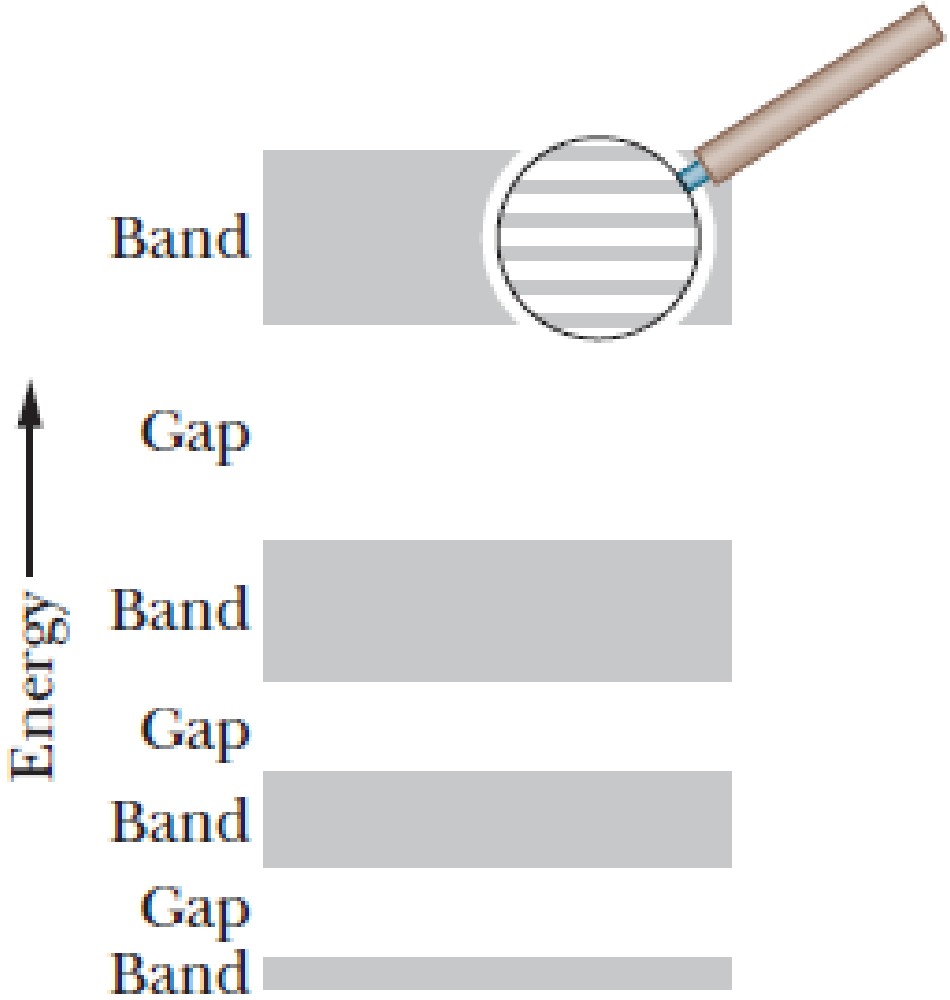
Quantised energy levels in atoms ⇒

energy <u>bands</u> in a solid

electron <u>cannot</u> have energy value within gaps

Number of electrons in a level is limited





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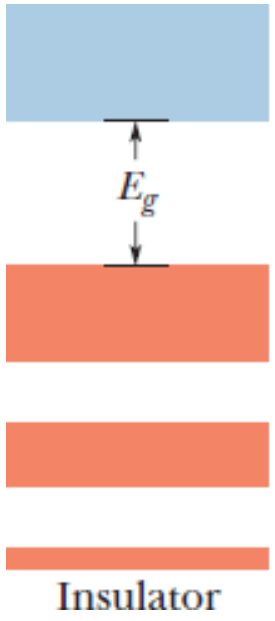
Insulators and Conductors

Copper
$n \approx 9 \times 10^{28}$ electrons/m^3

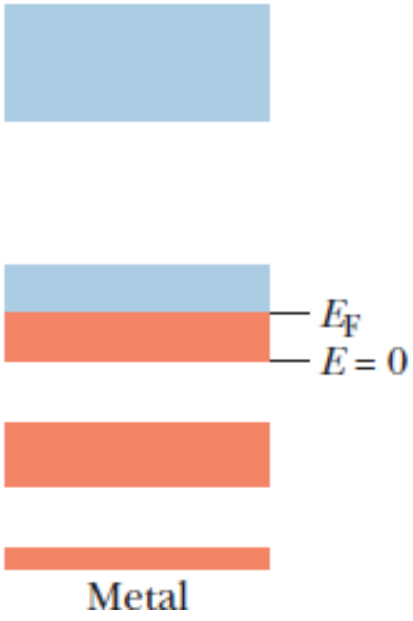
Electrons not free to move
High resistance
Examples
Plastic, Wood, Pure water

Conduction band



In an insulator, electrons need a big energy jump.

In a conductor, electrons need only a small energy jump.



Electrons free to move
Low resistance
Examples
Metals, impure water, human beings, ionized gas

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Semiconductors

Silicon
$n \approx 1x10^{16}$ electrons/m^3

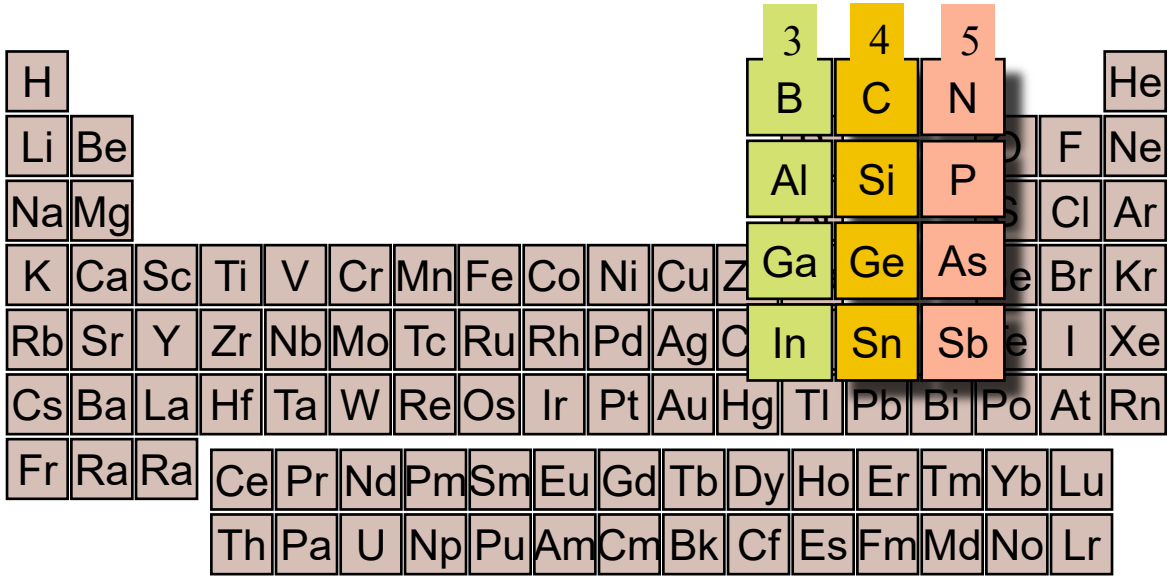
Insulators

- Relatively small gap between bands

Deliberate doping with trace elements provides a few electrons (or holes) free to move

Column 4 elements from the periodic table

- C, Si, Ge
- GaAs, CdTe
- InGaAs
- HgCdTe
- ZnS



Superconductors

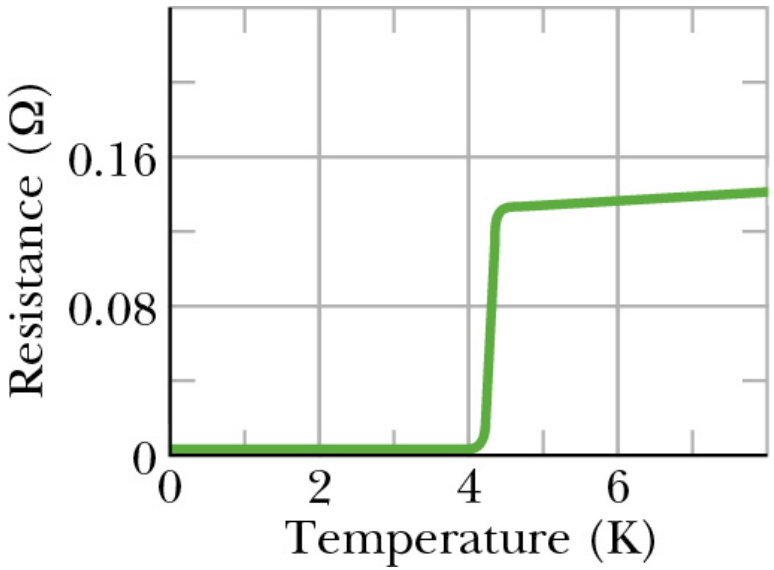
Insulators or poor conductors at normal temperatures

Resistance identically zero below a critical temperature, Tc

Examples

- Hg (Tc = 4K)
- Nb
- $YBa_2Cu_3O_7$ (Tc = 93K)
- BiSrCaCuO (Tc = 105K)

Mercury



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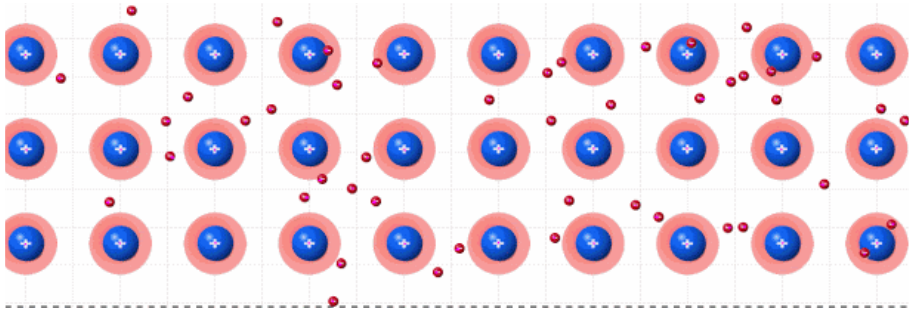
Electric Current

Before we can discuss the way capacitors (or anything else) behave in circuits, we need to look in depth at electric currents...

<u>Isolated conductor</u>

Random thermal motion of electron

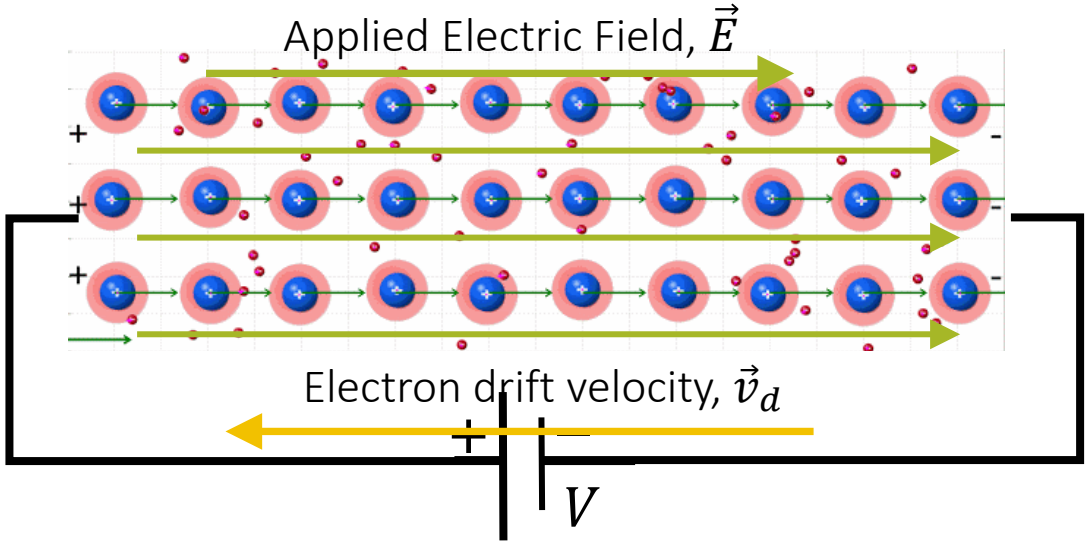
Speed $v \approx 10^6$ *m/s*



<u>Battery connected</u>

Random thermal motion of electron unchanged

Drift speed $v_d \approx 10^{-4}$ *m/s* superimposed



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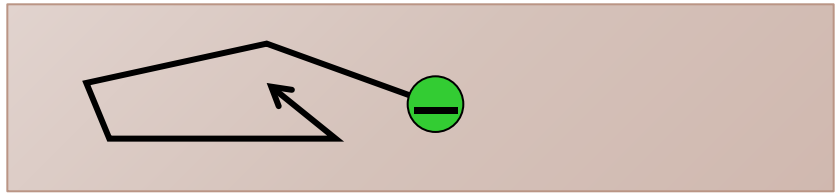
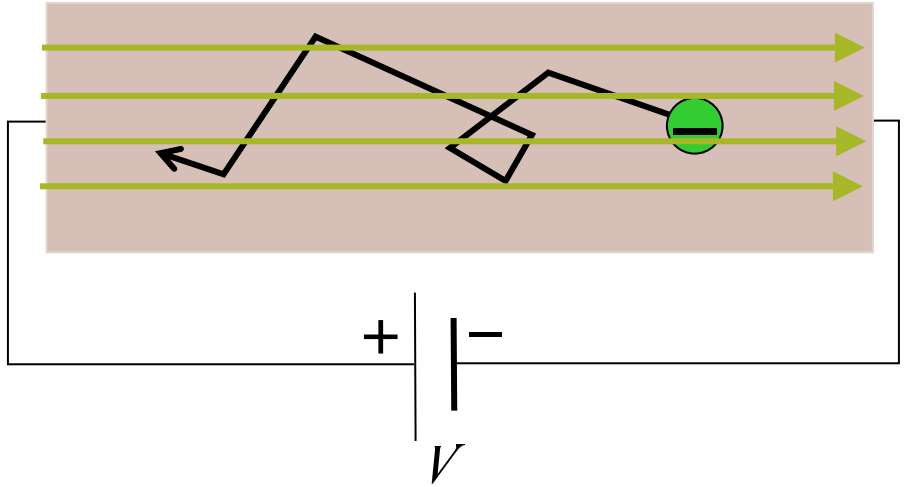
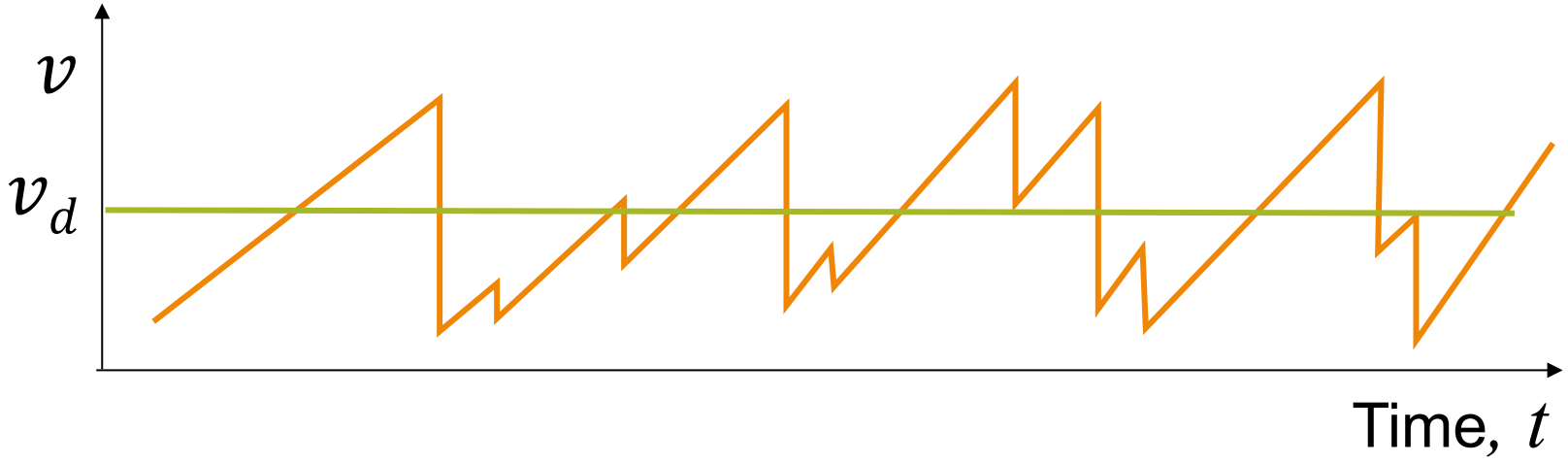




Electric Current

The electric field from the battery results in a force, and hence an acceleration



Electrons in the metal feel an acceleration from the electric field:

$$F = qE = ma \qquad\qquad a = qE/m$$

How much charge gets transported through the solid as a result?

Charge transport

Drift velocity transports charge

How much gets transported in a time dt?

- Distance travelled in time dt is $v_d dt$
- Area sweeps out element of length $v_d dt$

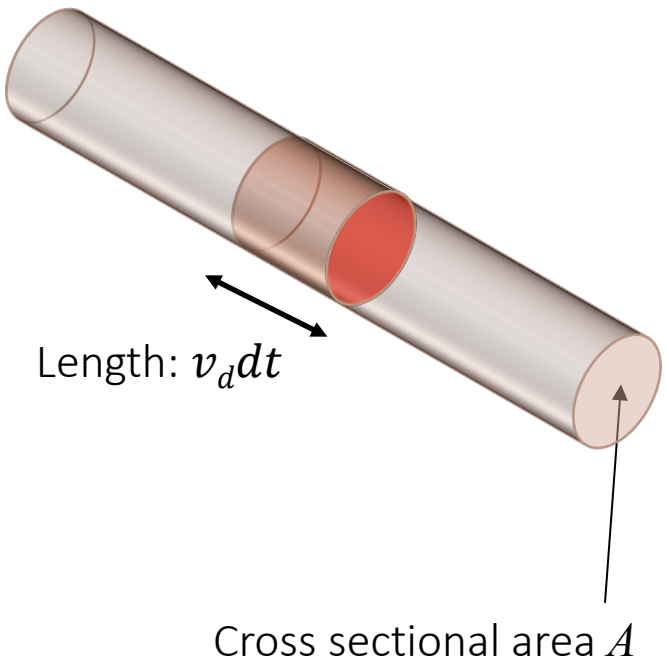
Volume of element

- $A \times v_d\, dt$
- Number = $n \times (A v_d dt)$
- n = number density of electrons

If each electron carries charge e, charge transported across surface in time dt is

- $dq = e \times n A v_d dt$

Same as amount of free charge in volume swept by A



$$dq = enAv_d dt$$

Electric current

Define the electric current

$$i = \frac{dq}{dt}$$

Time rate of transport of charge

$$i = \frac{dq}{dt} = enAv_d$$

Unit of current is Coulomb/second

- 1 Ampere = 1 Coulomb / 1 second
- 1A = 1 C/s

Direction of current

Defined to be direction of flow of positive charge

- Historic convention

Example:

A current of 10 Amperes flows in a 1 mm^2 copper wire. What is v_d given the number density of Cu is $n=8.45\times 10^{22}$ atom/cm^3 ?

Ans: 7.4×10^{-2} cm/s

$i = enAv_d$

Current density

Can also define current density, $\vec{J}$

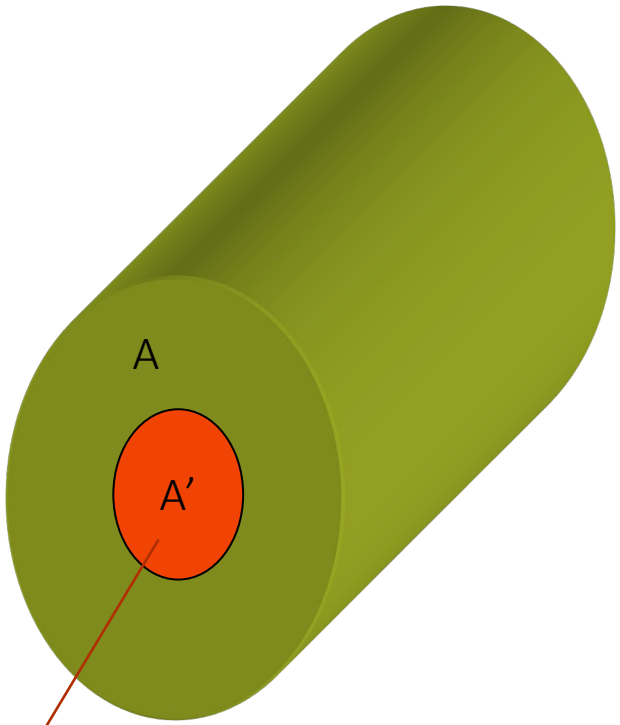
- Vector quantity in direction of charge flow
- Local (microscopic) viewpoint
- Units of current per unit area, A.m^{-2}

If a uniform current, i flows through a surface with area A,

$$J = \frac{i}{A} = e.n.v_d$$

Can also write

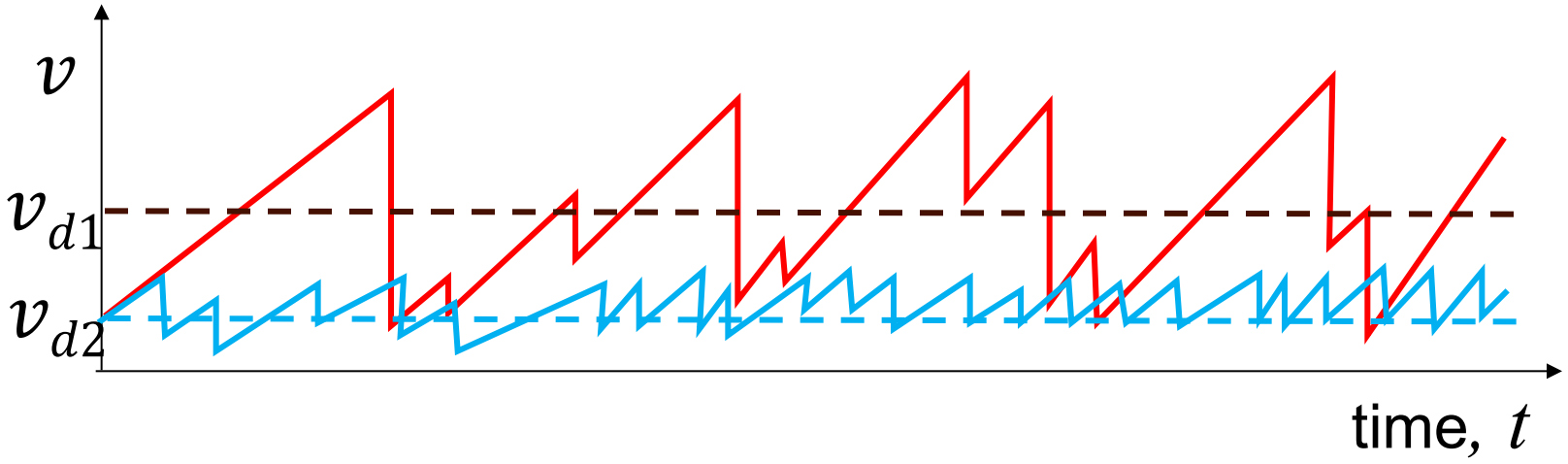
$$\vec{J} = e.n.\vec{v}_d$$



Current through here will be JA'

Define Resistance

Some materials cause more scattering of the electrons than others:

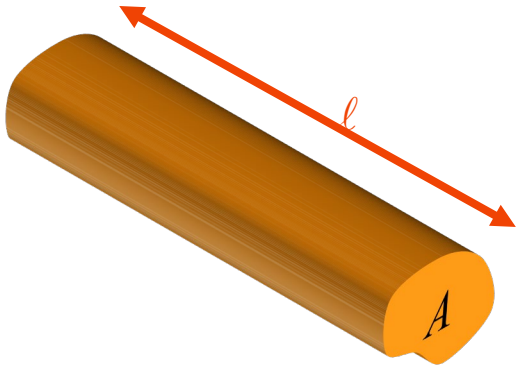


Here material 2 has more frequent scattering than material 1 so $v_{d2} < v_{d1}$

For the same electric field, less current flows in material 2:

$$i_1 = e.n.A.v_{d1}$$
$$i_2 = e.n.A.v_{d2} < i_1$$

Resistance and Ohm's Law



Consider various observations

$|\vec{E}| = V/\ell$ (uniform field)

$v_d \propto |\vec{E}| = V/\ell$

$i \propto A v_d$

So we write $i \propto \frac{A}{\ell} V$

Hence $i = \frac{A}{\rho \ell} V = \frac{V}{R}$

ρ and R constants

Introduce Ohm's Law

$$i = \frac{V}{R}$$

Resistance, R

Units: 1 ohm = 1 Ω = 1 V/A

Resistance of form

$$R = \rho \frac{\ell}{A}$$

ℓ is length of conductor

A is cross-sectional area of conductor

ρ is the resistivity of the material

Depends on material

A function of temperature and structure

Typically determined empirically

Units: Ω.m

Resistivity, ρ

Small ρ

- good conductor

Large ρ

- poor conductor
- good insulator

Function of temperature

ρ is a property of the material

- $R = \rho\, \ell / A$ gives resistance for a particular piece
- R depends on geometry and material

Material	Resistivity, ρ (Ω m)
Copper	1.7×10^{-8}
Tungsten (20°C)	5.6×10^{-8}
Tungsten (1500°C)	5.0×10^{-7}
Iron	9.7×10^{-8}
Nichrome	1.5×10^{-6}
Seawater	0.22
Blood (average)	1.6
Muscle	13
Fat	25
Pure water	2.4×10^{5}
Cell membrane	3.6×10^{7}

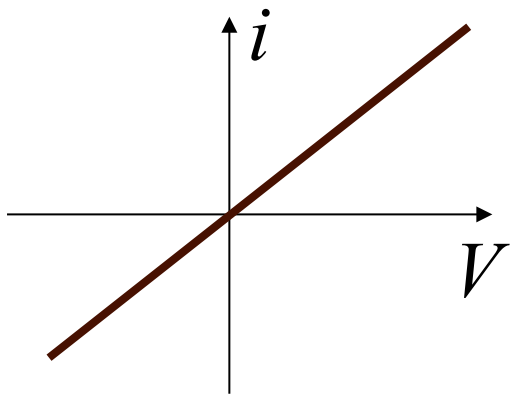
Ohm's Law

If a material or device obeys Ohm's law then

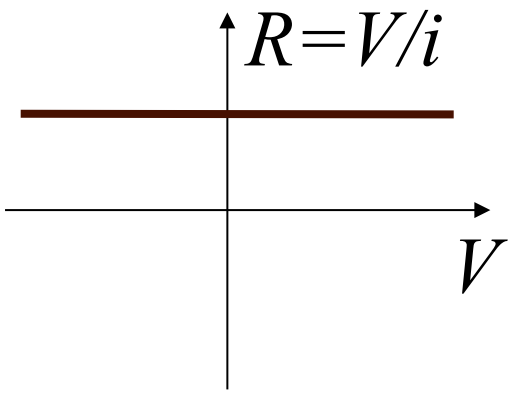
$$V = iR$$

and R is independent of i and V

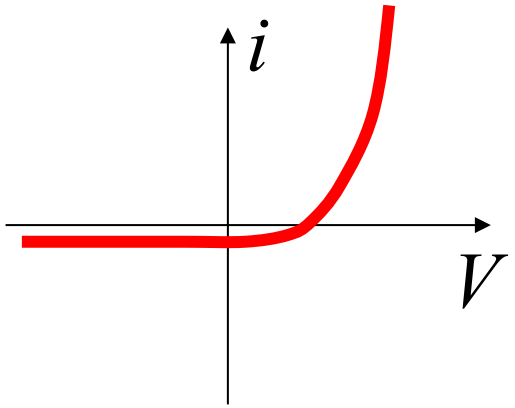
Carbon film resistor



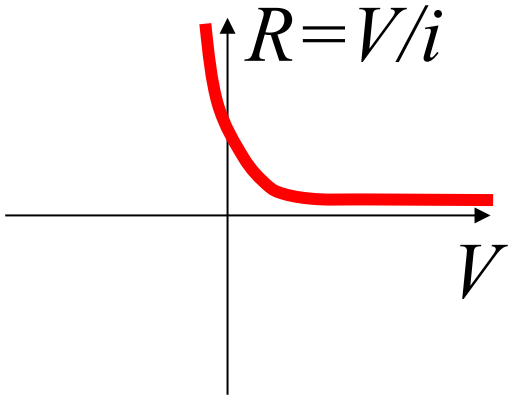
This device obeys Ohm's Law



Diode



This device does not!



Summary and Further Work

KEY OUTCOMES

You should be able to …

- Briefly explain the differences between conductors and insulators at the atomic scale
- Explain the appearance of an electric current on the microscopic scale
- Relate this to measurable currents
- Calculate currents and current densities
- Explain the concept of resistance
- Explain the origins of and use Ohm's Law

FURTHER WORK

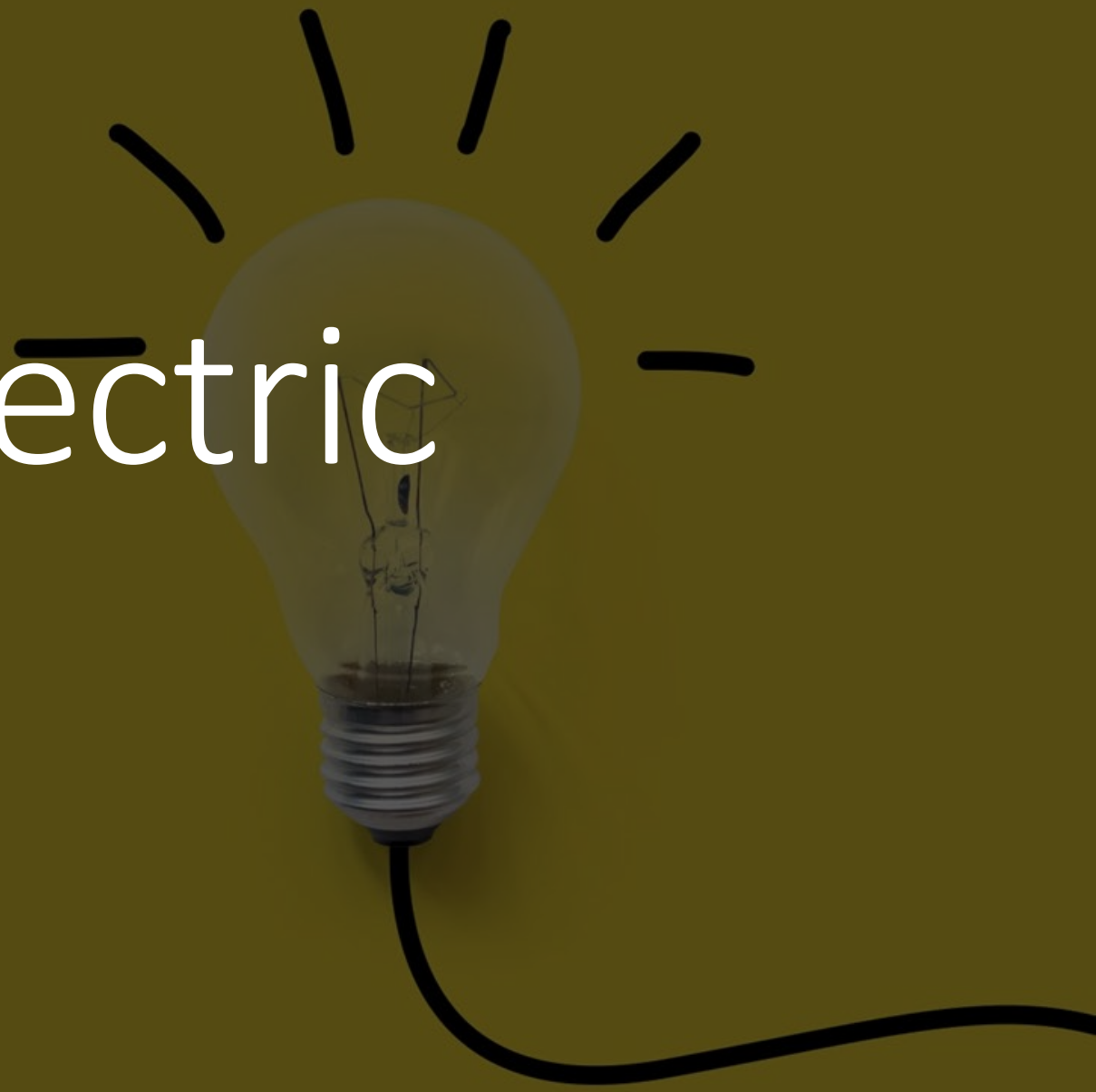
Reading

- Chapter 26
  - *Sections 26.1-4*

Problems

- Problems 26.2, 26.5, 26.16, 26.28

Power and electric circuits



Key concepts

PREVIOUS TOPIC

Microscopic origins of electric current

Current and current density

Resistance

Ohm’s Law

THIS TOPIC

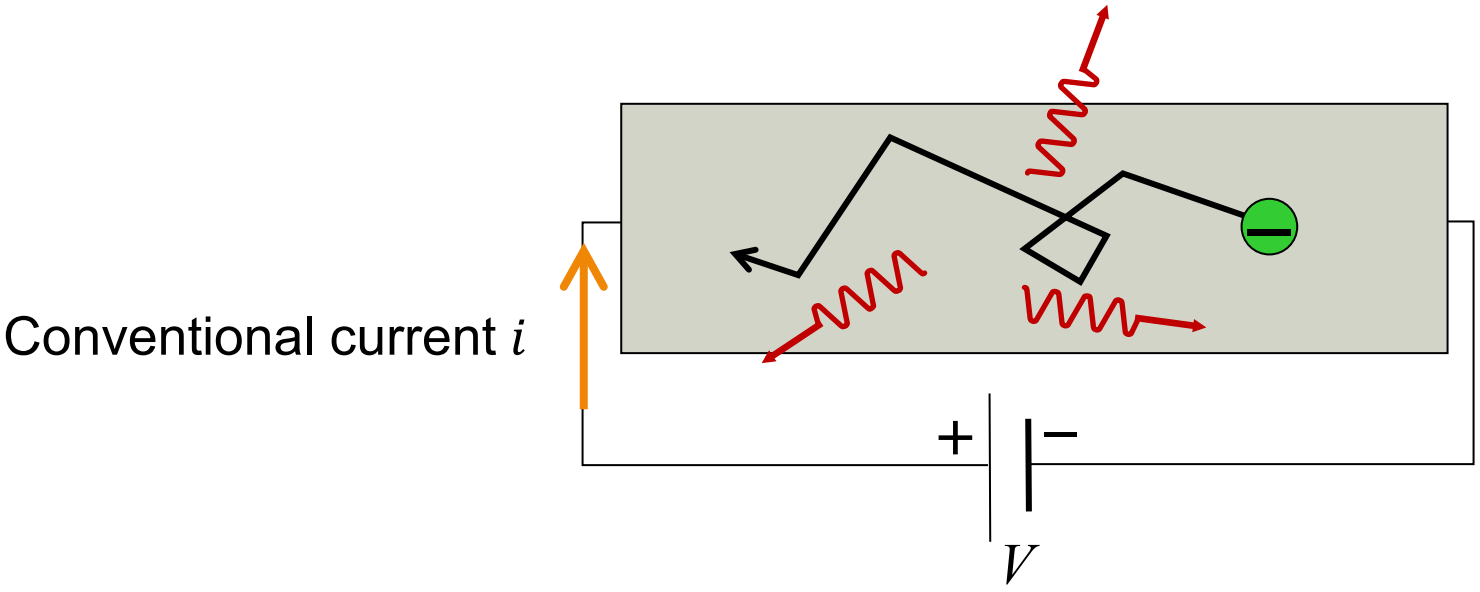
Power in electric circuits

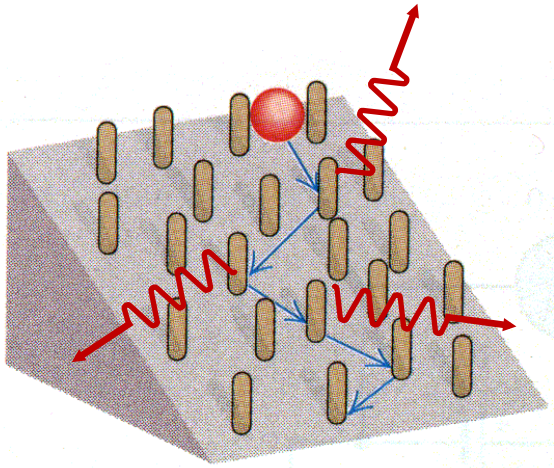
Single loop circuits

Power and Electric circuits

Power: Energy is transferred to the material by scattering electrons

The material gets hot!





University Physics, Young and Freedman, Addison-Wesley © 2000

Power and Electric circuits

Energy must be fed into the system!

Consider charge dq moving through potential difference V

Work done by field is Vdq

Power transferred to device = Rate at which work is done by the electric field

$$P = \frac{Vdq}{dt} = V\frac{dq}{dt} = iV$$

Power transferred to a device: $P = iV$

Unit: Joules/second = Watts, W

For a resistor power dissipated: $P = i \times iR = i^2R$

Powerlines

Why are power lines at very high voltages (e.g. 500kV) when household circuits run at 230V?

Example: LaTrobe power station, P=1200 MW

Copper cable, 2cm diameter, 20km long

Resistance approx. 1Ω

If power was transmitted at 230V, power dissipated in cable

$$P_{loss} = 2.7 \times 10^{13}\ \text{W}$$

More than the entire power station!

Use V=500kV for most of distance

$P_{loss} = 6 \times 10^{6}$ W = 0.5% of 1200MW

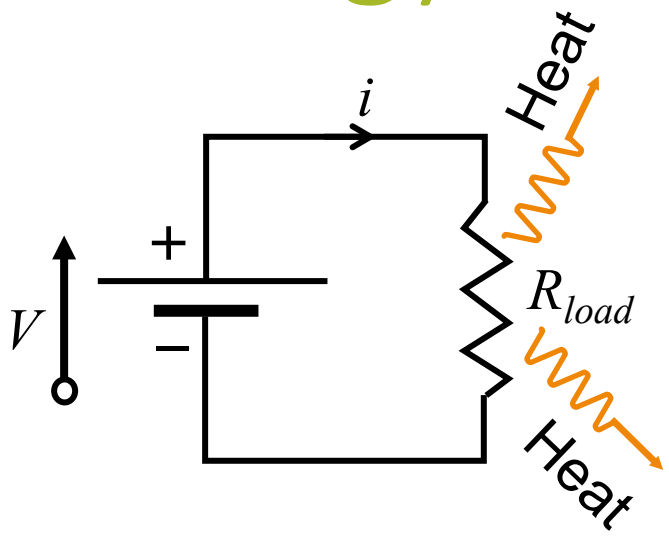
Image: sa.gov.au dtsqld.com.au

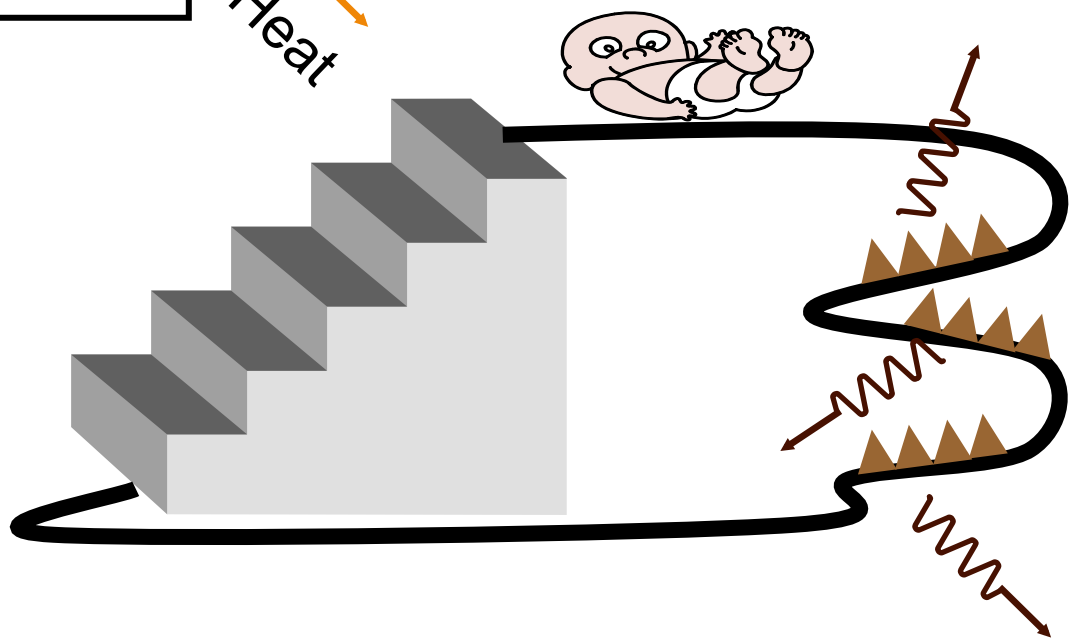
Circuit Theory and Energy

A battery is a source of electromotive force (emf), or potential

This drives currents around circuits

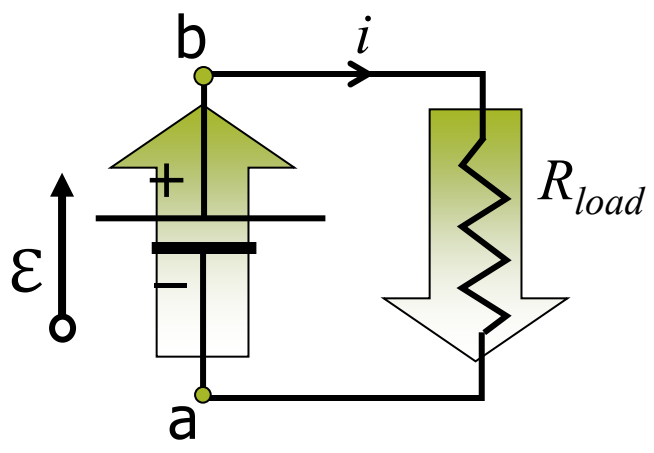
In our gravity analogy, friction dissipates the kinetic energy, the escalators take the place of the emf





Single Loop circuits

Total change in electric potential around a closed loop is zero



Potential increase going from a → b

$$V_b - V_a = \mathcal{E}$$

$\mathcal{E}$ is the emf (potential) provided by the battery

Potential decrease going from b → a

$$V_a - V_b = -iR_{load}$$

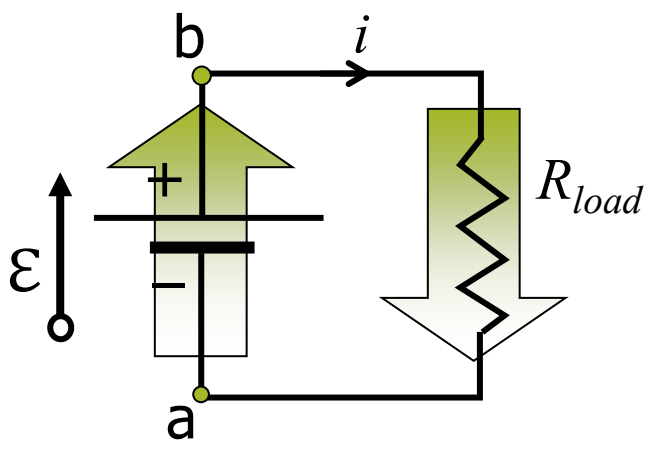
So

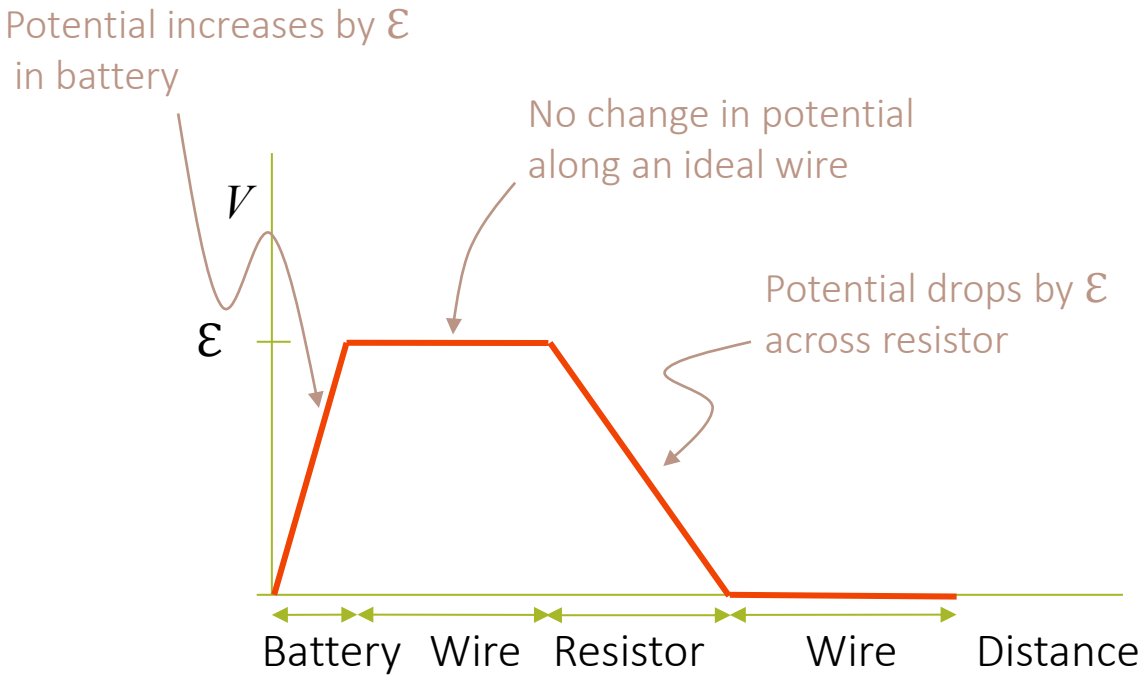
$$\varepsilon - iR_{load} = 0$$

i.e.

$$\boxed{i = \frac{\mathcal{E}}{R_{load}}}$$

The single-resistor circuit





Summary and Further Work

KEY OUTCOMES

You should be able to …

- Calculate changes in potential in a single-loop circuit
- Calculate power delivery and dissipation in circuit elements
- Determine the current flowing through elements in single loop circuits

FURTHER WORK

Reading

- Chapter 26
  - *Sections 26.5*

Problems

- Problems 26.48, 27.3

Kirchhoff’s Rules

Key concepts

PREVIOUS TOPIC

Power in electric circuits

Single loop circuits

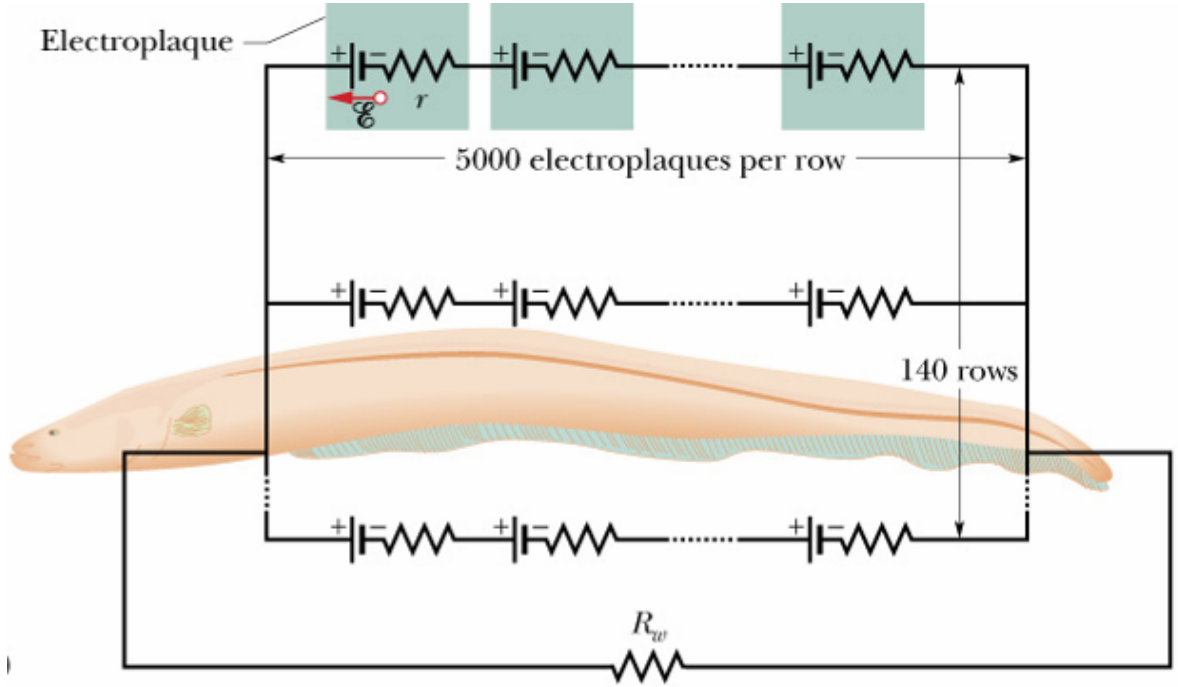
THIS TOPIC

Kirchhoff's Rules

Application to multi-loop circuits

More complex circuits

The equivalent circuit of an electric eel



Have multiple loops containing batteries, resistors and perhaps even capacitors

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Kirchhoff's Laws

Nothing new here…

Loops must have a net potential change of zero

$$\Delta V_{loop} = \sum_n \Delta V_n = 0$$

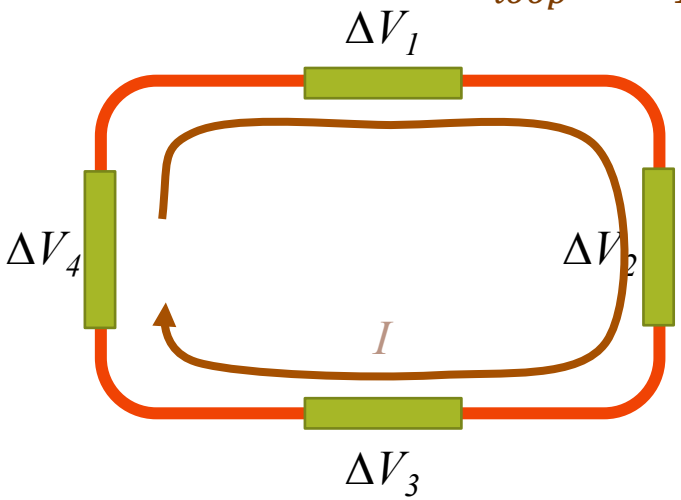
- Traverse battery in direction of emf +
- Traverse battery opposite to emf –
- Traverse resistor in direction of current -
- Traverse resistor opposite to current +

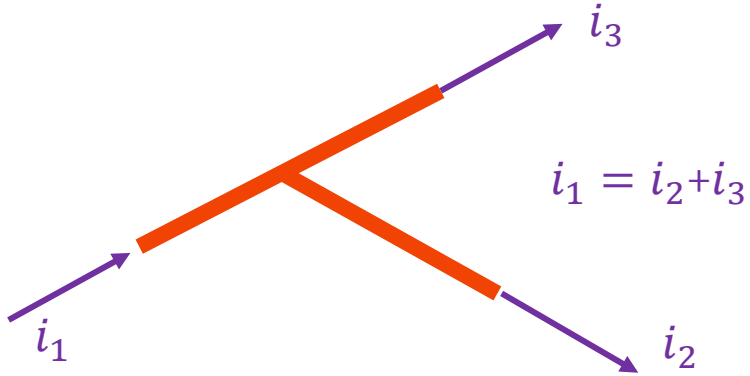
Current (the flow of charge) is conserved at junctions

$$\sum_n i_n = 0$$

This is all about logic

$\Delta V_{loop} = \Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4$





Resistors in series

Resistances add

Current is the same through each

If not, where would charges go?

The potential drop across each resistor

Given by Ohm's Law

$$+\varepsilon - iR_1 - iR_2 = 0 \quad \Rightarrow \quad \varepsilon = i(R_1 + R_2)$$

Equivalent circuit

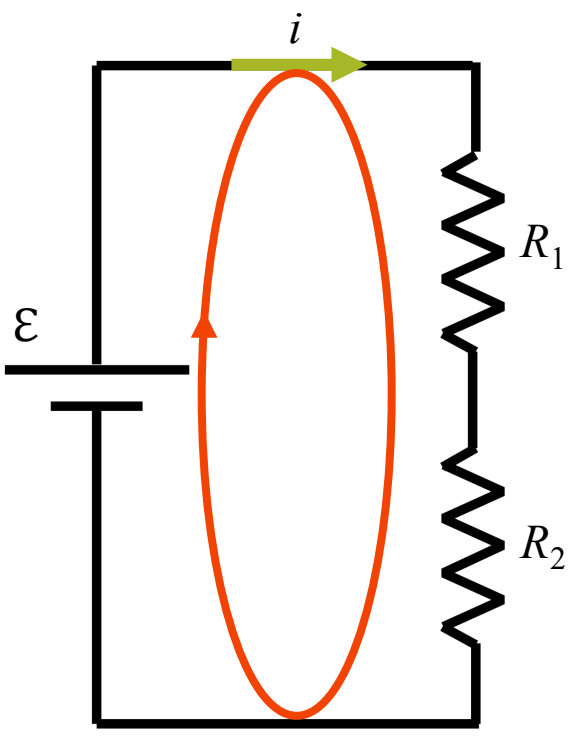
$$+\varepsilon - iR_T = 0 \quad \Rightarrow \quad \varepsilon = \mathrm{i}R_\mathrm{T}$$

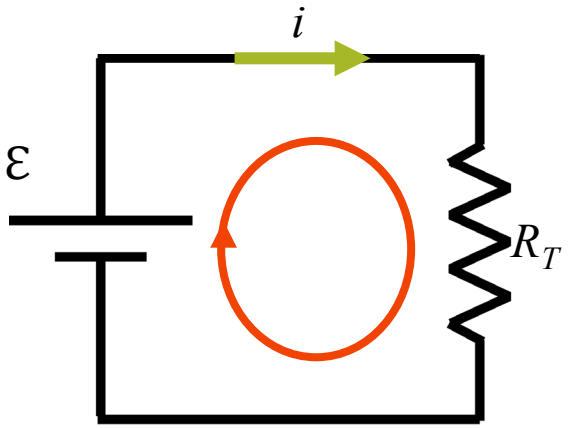
Hence

$$R_T = R_1 + R_2$$

In general

$$R_T = \sum_n R_n$$





Resistors in parallel

Each resistor has a potential drop of V_{bat}

The current in each resistor is given by Ohm's Law

The total current from the battery is the sum $i = i_1 + i_2$

Loop 1: $+\mathcal{E} - i_1 R_1 = 0 \implies i_1 = \mathcal{E}/R_1$

Loop 2: $+\mathcal{E} - i_2 R_2 = 0 \implies i_2 = \mathcal{E}/R_2$

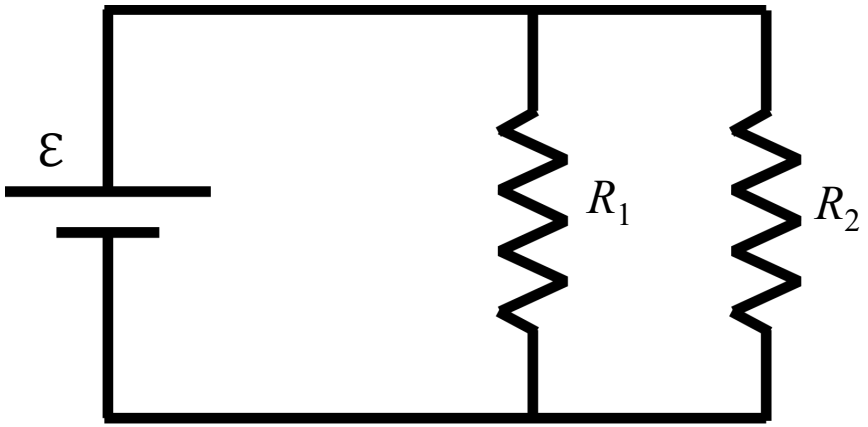
Current rule: $i = i_1 + i_2 = \mathcal{E}/R_T = \mathcal{E}/R_1 + \mathcal{E}/R_2$

Hence

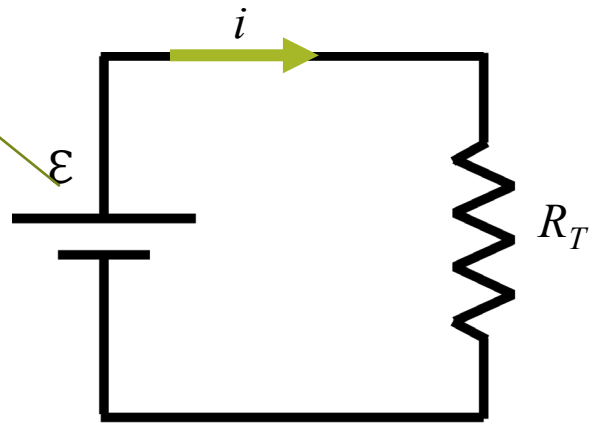
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

General result

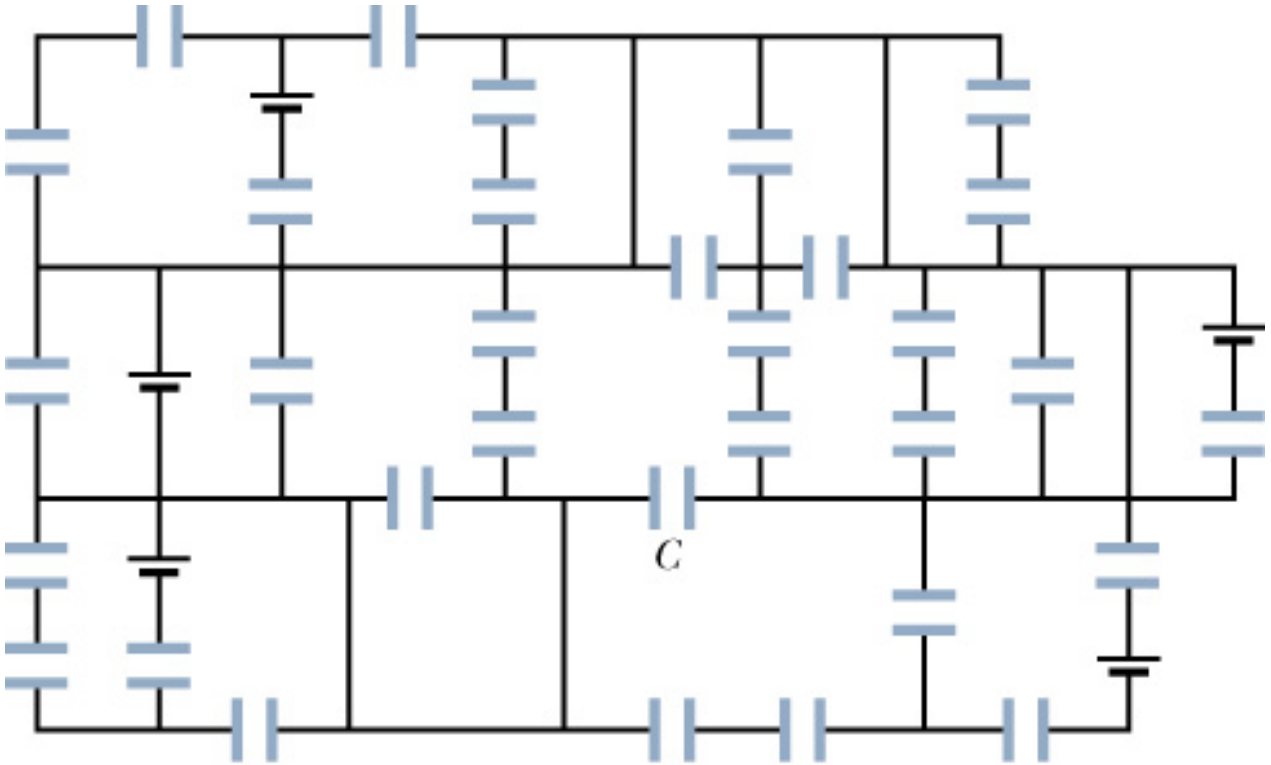
$$\frac{1}{R_T} = \sum_n \frac{1}{R_n}$$



Batteries have fixed potential not current



Cap-monster maze



All the capacitors have a have a capacitance of 6.0 μF and all the (ideal) batteries have an emf of 10 V. What is the charge on capacitor C?

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Summary

Kirchhoff's Rules for circuit analysis allow the voltage and current to be calculated at any point in a complex circuit

Rule 1: Conservation of charge

At a node

$$\sum_n i_n = 0$$

Rule 2: Conservation of energy

Around a loop

$$\Delta V_{loop} = \sum_n \Delta V_n = 0$$

Summary and Further Work

KEY OUTCOMES

You should be able to …

- Explain Kirchhoff's Rules in physical terms
- Apply Kirchhoff's Rules to multi-loop circuits

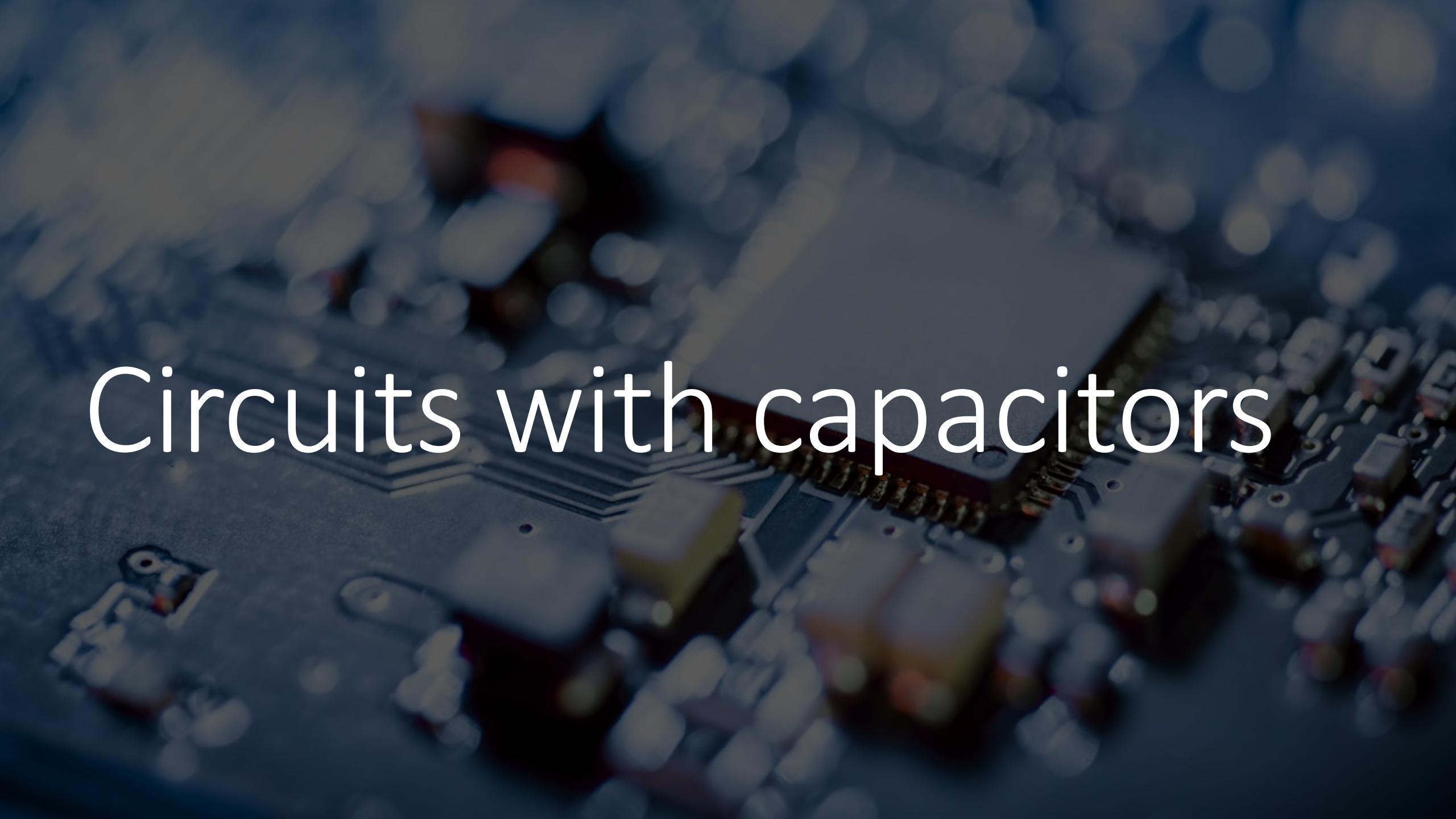
FURTHER WORK

Reading

- Chapter 27
  - *Sections 27.1-2*

Problems

- Problems 27.10, 27.15, 27.21, 27.26, 27.29



Circuits with capacitors

Key concepts

PREVIOUS TOPIC

Kirchoff's Rules

Application to multi-loop circuits

THIS TOPIC

Circuits containing capacitors

Time-dependent current

Summary

Kirchhoff's Rules for circuit analysis allow the voltage and current to be calculated at any point in a complex circuit

Rule 1: Conservation of charge

At a node

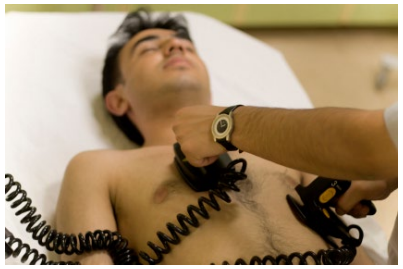
$$\sum_n i_n = 0$$

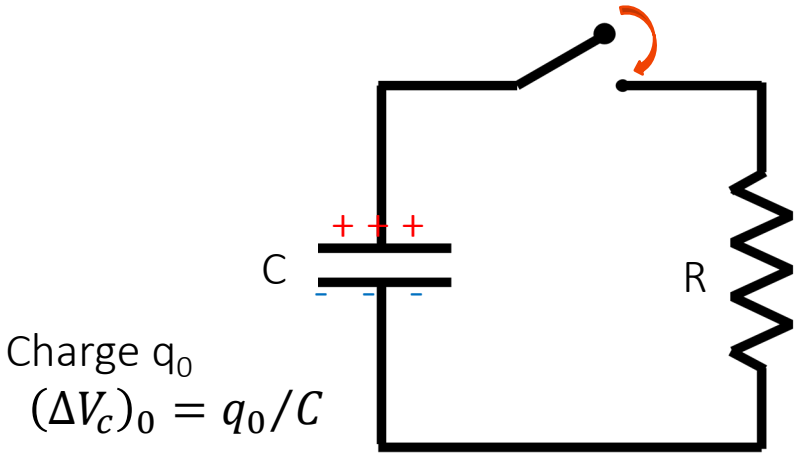
Rule 2: Conservation of energy

Around a loop

$$\Delta V_{loop} = \sum_n \Delta V_n = 0$$

Capacitors in circuits





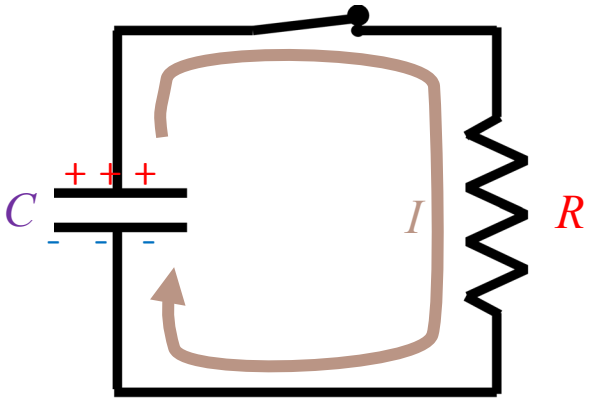
Consider how a capacitor discharges

We connect a charged capacitor and resistor in series

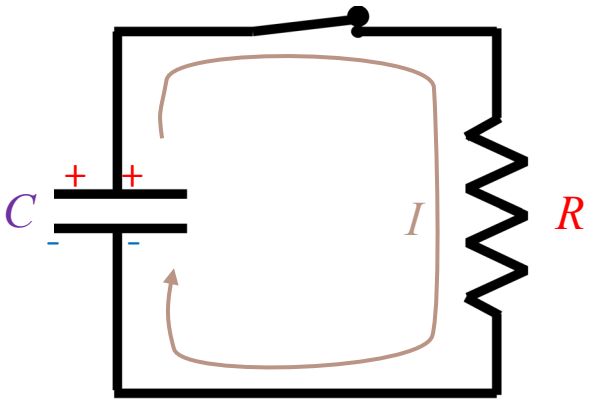
The switch closes at t = 0

The discharging *RC* circuit

Immediately after the switch closes



At a later time



After a sufficiently long time, the capacitor is uncharged and there is zero current in the circuit

Discharging capacitor

Current at time t

$$i(t) = \frac{\Delta V_C(t)}{R}$$

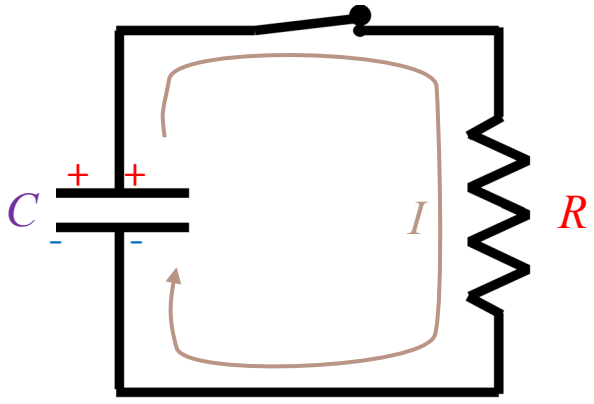
Reduced charge on capacitor results in reduced potential difference across capacitor

$$\Delta V_C(t) = \frac{q(t)}{C}$$

Replace $i(t)$ with $-\frac{dq(t)}{dt}$

Replace $\Delta V_C(t) = \frac{q(t)}{C}$

We have a *differential equation*:



$$\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{RC} = 0$$

Discharging capacitor

Homework: Substitute the answer for the charge as a function of time, $q(t)$, back into the differential equation and *prove* that it is a valid solution.

Need to solve $\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{RC} = 0$ with $q(0) = q_0$

You don't need to be able to solve it. But it may be instructive to see it done once…

$$\frac{\mathrm{d}q}{q} = -\frac{1}{RC}\mathrm{d}t$$

$$\therefore \int_{q_0}^{q(t)} \frac{\mathrm{d}q'}{q'} = -\frac{1}{RC}\int_0^t \mathrm{d}t'$$

$$\therefore \ln q\,(t) - \ln q_0 = \ln\left(\frac{q(t)}{q_0}\right) = -\frac{t}{RC}$$

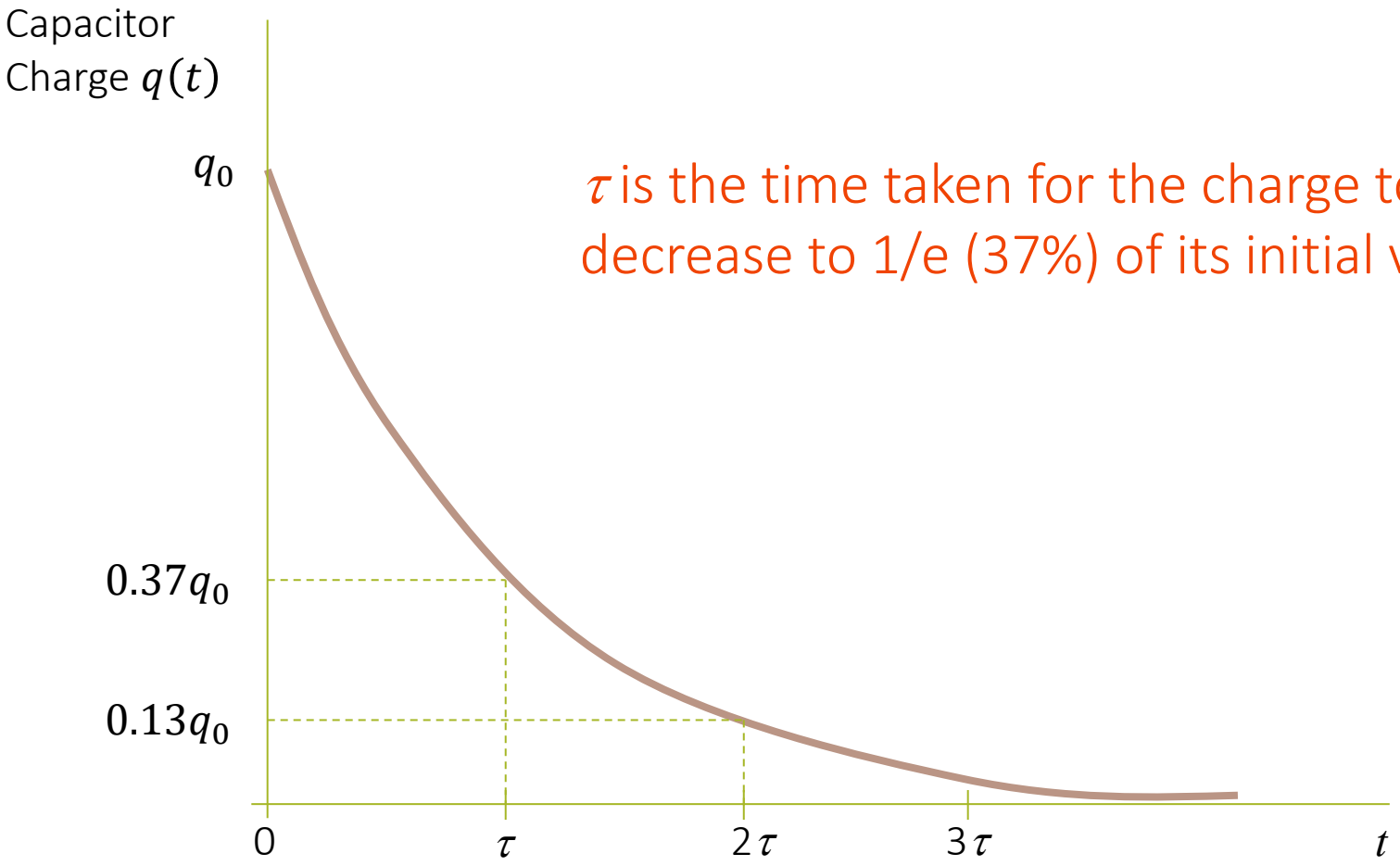
$$\therefore q(t) = q_0 e^{-t/\tau}, \text{ where } \tau = RC$$

$\tau = RC$ is the *time constant* of the circuit

Units: seconds (s)

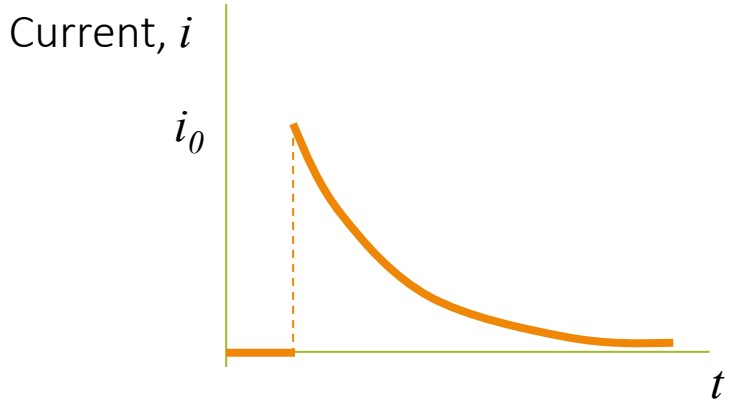
The time constant, $\tau = RC$

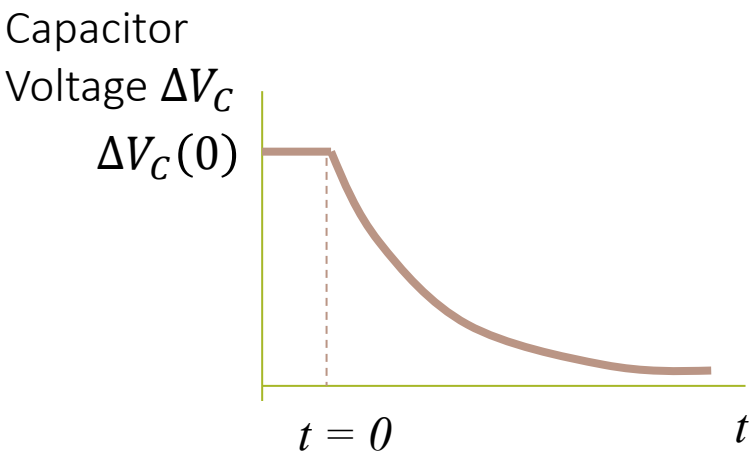
Capacitor charge
$q(t) = q_0 e^{-t/\tau}$



τ is the time taken for the charge to decrease to 1/e (37%) of its initial value

The discharging RC circuit





Capacitor charge - $q(t) = q_0 e^{-t/\tau}$

$\tau = RC$ is *time constant*

What are $\Delta V_C(t)$ and $i(t)$?

$q = C\Delta V_C(t)$ gives us $\Delta V_C(t)$

$$\Delta V_C(t) = (\Delta V_c)_0 e^{-t/\tau} = \frac{q_0}{C} e^{-t/\tau}$$

Exponential dependence

Differentiation yields:

$$i(t) = -\frac{dq(t)}{dt} = i_0 e^{-t/\tau} = \frac{q_0}{RC} e^{-t/\tau}$$

Current and voltage in circuit decay exponentially

Capacitor charging

Initially uncharged capacitor

Switch closes at $t = 0$

Initial conditions: at $t = 0, q = 0$

Kirchhoff 2:

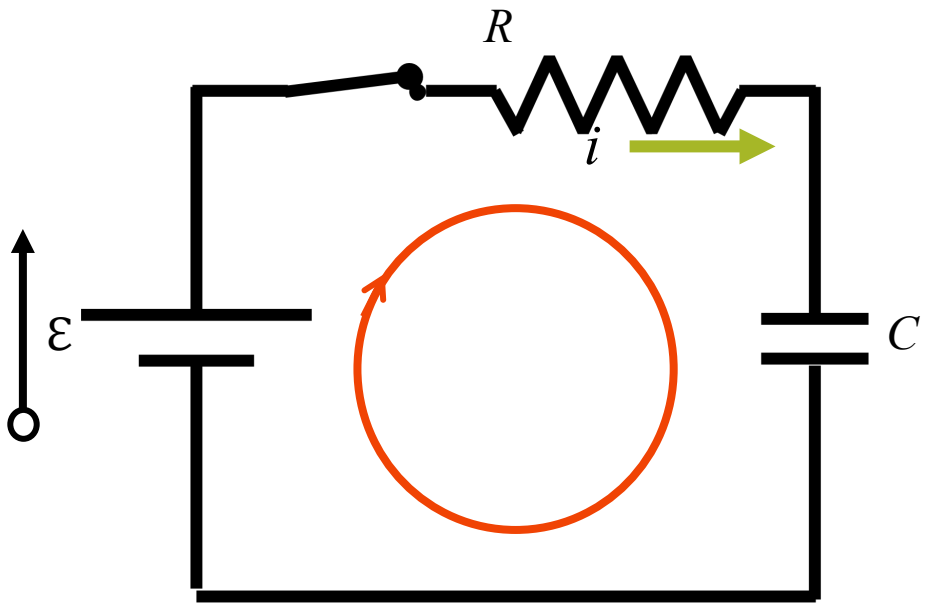
$$+\mathcal{E} - iR - \frac{q}{C} = 0$$

Solution is:

$$q(t) = C\mathcal{E}(1 - e^{-t/(RC)})$$

and

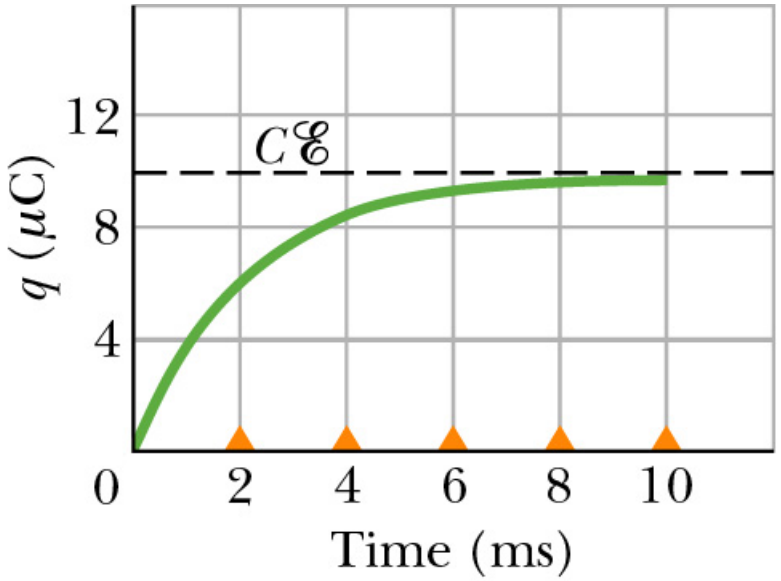
$$i(t) = \frac{dq}{dt} = \frac{\mathcal{E}}{R} e^{-t/(RC)}$$



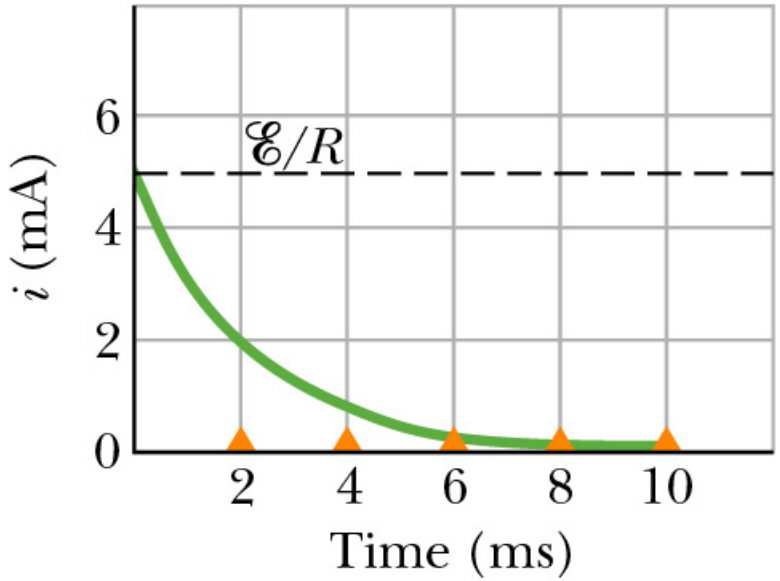
Define $\tau = RC$ = time constant unit: seconds, s (time for 63% full)

Capacitor charging

$$q(t) = C\mathcal{E}(1 - e^{-t/(RC)})$$



$$i(t) = \frac{dq}{dt} = \frac{\mathcal{E}}{R} e^{-t/(RC)}$$



$C\mathcal{E} = 10\mu C$

$\mathcal{E}/R = 5mA$

$RC = 2ms$

$R = 2000\Omega$

$C = 1\mu F$

$\mathcal{E} = 10V$

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Charge, discharge cycles…

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Summary and Further Work

KEY OUTCOMES

You should be able to …

- Analyse circuits including capacitors
- Apply Kirchhoff's Rules more broadly

FURTHER WORK

Reading

- Chapter 27
  - *Sections 27.3*

Problems

- Problems 27.57, 27.59

Course Outline

1. Electrostatics
   - Stationary charged particles
   - Electrostatic forces and fields
   - Gauss' Law
2. Energy and potential
   - Electric potential energy
   - Electric potential
   - Capacitance
3. Circuit Theory
   - Electrons moving in wires
   - Resistors, capacitors, batteries
4. Magnetism
   - Moving charged particles
   - Magnetic fields, forces, materials
   - Magnetic induction

Finished

Next!